

ADAPTIVE MULTI CELL BATTERY

STATUS – HEALTHY - WOKWI

The Wokwi simulation interface displays the following code on the left and a circuit diagram on the right. The circuit diagram shows a battery pack with four cells, a microcontroller, and a Blynk module. The code is as follows:

```
172...
215 else {
216   Blynk.virtualWrite(V6, health);
217 }
218
219 if (equalCells) {
220   Blynk.virtualWrite(V7, "None");
221   Blynk.virtualWrite(V8, "None");
222 }
223 else {
224   String weakStr = "";
225   for (int i = 0; i < weakCount; i++) {
226     weakStr += String(weakCells[i]);
227     if (i < weakCount - 1) weakStr += ",";
228   }
229
230   String strongStr = "";
231   for (int i = 0; i < strongCount; i++) {
232     strongStr += String(strongCells[i]);
233     if (i < strongCount - 1) strongStr += ",";
234   }
235
236   Blynk.virtualWrite(V7, weakStr);
237   Blynk.virtualWrite(V8, strongStr);
238 }
239
240 // SERIAL
241 Serial.println("\n=====");
242
243 for (int i = 0; i < 4; i++) {
244   Serial.print("Cell ");
245   Serial.print(i + 1);
246   Serial.print(": ");
247   Serial.println(cell[i]);
248 }
```

The simulation window shows a circuit diagram with a battery pack, a microcontroller, and a Blynk module. The Blynk module displays the following data:

- Cell 1: 2.76
- Cell 2: 2.80
- Cell 3: 2.76
- Cell 4: 2.76
- Weak: 1,3,4
- Strong: 2
- Health: Healthy

STATUS – HEALTHY - BLYNK

The Blynk dashboard displays the following data:

- CELL1**: 2.76V
- CELL2**: 2.8V
- CELL3**: 2.76V
- CELL4**: 2.76V
- AVERAGE VOLTAGE**: 2.77V
- IMBALANCE PERCENTAGE**: 1.44%
- HEALTH**: Healthy
- WEEKCELL**: 1,3,4
- STRONGCELL**: 2

STATUS –MINOR- WOKWI

The Wokwi IDE interface displays a C++ sketch for a battery management system. The code includes a `batteryEngine` function that iterates through four cells, printing their voltage and health status. The `setup` function initializes the serial port, pins, LCD, and Blynk module. The `loop` function runs the Blynk module and the battery engine timer.

```
136 void batteryEngine() {  
137   Serial.print("Weak: ");  
138   for (int i = 0; i < weakCount; i++) {  
139     Serial.print(weakCells[i]);  
140     if (i < weakCount - 1) Serial.print(",");  
141   }  
142   Serial.println();  
143   Serial.print("Health: ");  
144   Serial.println(health);  
145   Serial.println("=====");  
146 }  
147  
148 void setup() {  
149   Serial.begin(115200);  
150   pinMode(LED_PIN, OUTPUT);  
151   pinMode(BUZZER, OUTPUT);  
152   pinMode(RELAY, OUTPUT);  
153   lcd.init();  
154   lcd.backlight();  
155   Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);  
156   timer.setInterval(2000L, batteryEngine);  
157 }  
158  
159 void loop() {  
160   Blynk.run();  
161   timer.run();  
162 }
```

The simulation window shows a circuit diagram with a battery pack and a display. The display shows the average voltage and health status. The console output shows the following data:

Cell	Voltage
Cell 1	2.76
Cell 2	2.80
Cell 3	2.82
Cell 4	2.76

Additional data shown in the console:

- Strong: 3
- Weak: 1,4
- Health: Minor

STATUS –MINOR- BLYNK

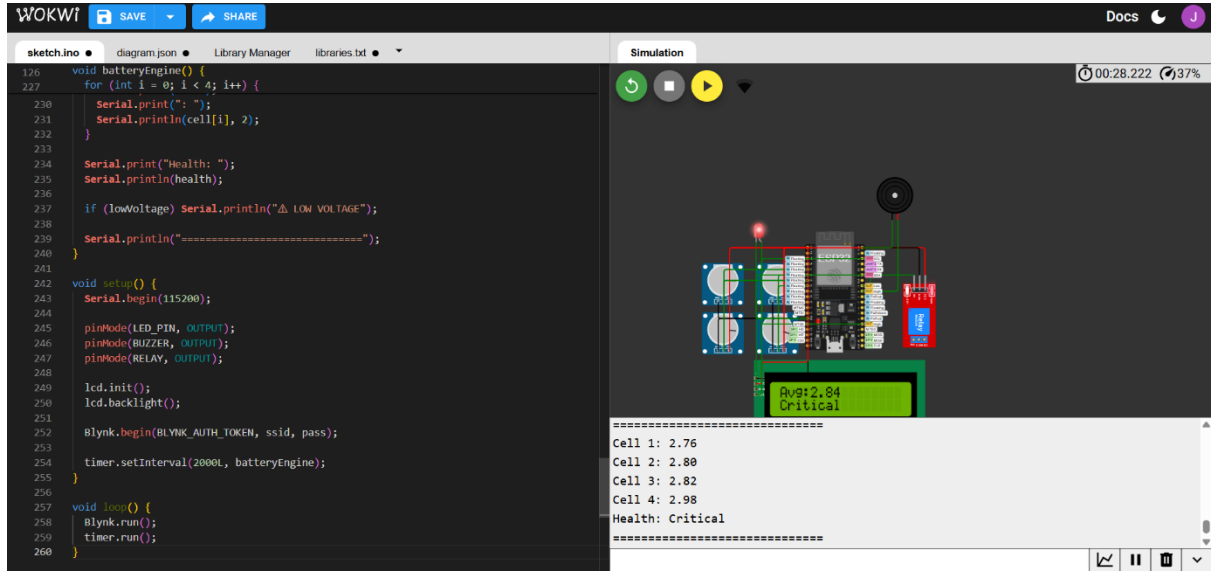
The Blynk Console dashboard for the 'Adaptive multicell battery' project shows the following data:

Cell	Voltage
CELL1	2.76V
CELL2	2.8V
CELL3	2.82V
CELL4	2.76V

Summary Metrics:

- AVERAGE VOLTAGE: 2.79V
- IMBALANCE PERCENTAGE: 2.2%
- HEALTH: Minor
- WEEKCELL: 1,4
- STRONGCELL: 3

STATUS –CRITICAL- WOKWI



```
126 void batteryEngine() {
127   for (int i = 0; i < 4; i++) {
128     Serial.print(i);
129     Serial.print(" ");
130     Serial.println(cell[i], 2);
131   }
132   Serial.print("Health: ");
133   Serial.println(health);
134   if (lowVoltage) Serial.println("⚠ LOW VOLTAGE");
135   Serial.println("=====");
136 }
137
138 void setup() {
139   Serial.begin(115200);
140   pinMode(LED_PIN, OUTPUT);
141   pinMode(BUZZER, OUTPUT);
142   pinMode(RELAY, OUTPUT);
143   lcd.init();
144   lcd.backlight();
145   Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
146   timer.setInterval(2000L, batteryEngine);
147 }
148
149 void loop() {
150   Blynk.run();
151   timer.run();
152 }
```

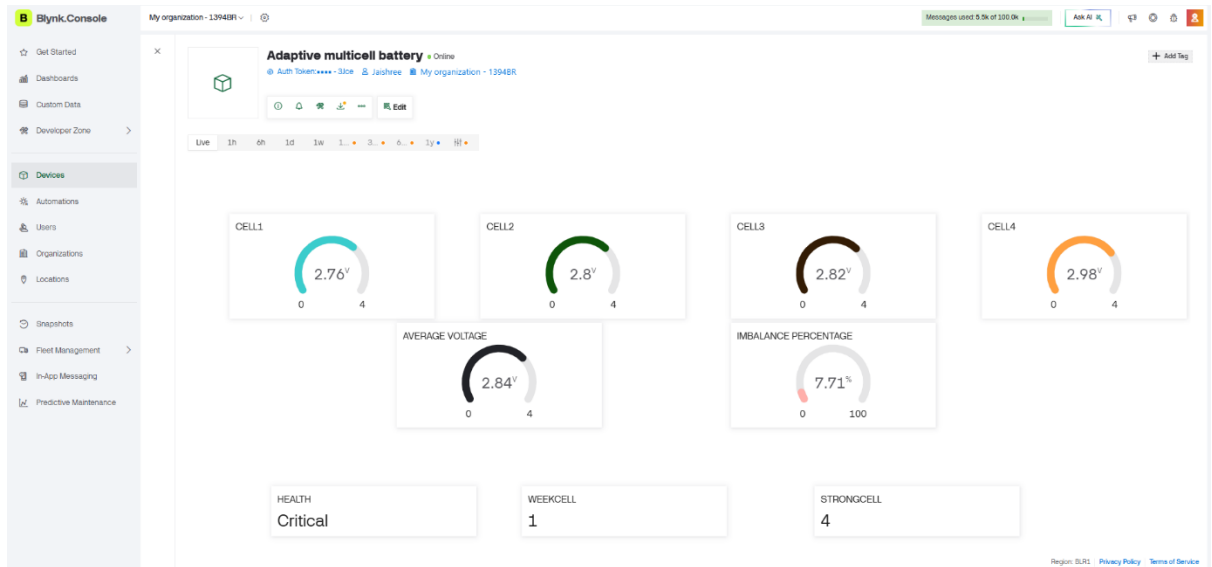
Simulation

00:28.222 37%

Avg: 2.84
Critical

Cell 1: 2.76
Cell 2: 2.80
Cell 3: 2.82
Cell 4: 2.98
Health: Critical
=====

STATUS –CRITICAL- BLYNK



Blynk Console

My organization - 13948R

Messages used 5.5k of 100.0k

Auth Token: ****-3ice jashree My organization - 13948R

Live 1h 6h 1d 1w 1m 3d 6d 1y H

CELL1: 2.76V

CELL2: 2.8V

CELL3: 2.82V

CELL4: 2.98V

AVERAGE VOLTAGE: 2.84V

IMBALANCE PERCENTAGE: 7.71%

HEALTH: Critical

WEEKCELL: 1

STRONGCELL: 4

Region: EU1 Privacy Policy Terms of Service

STATUS –FAILURE- WOKWI

The Wokwi simulation interface displays a circuit diagram of a battery pack with four cells, a buzzer, and a relay. The code editor shows the following code:

```
126 void batteryEngine() {
127   for (int i = 0; i < 4; i++) {
128     Serial.print(": ");
129     Serial.println(cell[i], 2);
130   }
131
132   Serial.print("Health: ");
133   Serial.println(health);
134
135   if (lowVoltage) Serial.println("⚠ LOW VOLTAGE");
136   Serial.println("=====");
137 }
138
139 void setup() {
140   Serial.begin(115200);
141
142   pinMode(LED_PIN, OUTPUT);
143   pinMode(BUZZER, OUTPUT);
144   pinMode(RELAY, OUTPUT);
145
146   Icd.init();
147   Icd.backlight();
148
149   Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
150   timer.setInterval(2000L, batteryEngine);
151 }
152
153 void loop() {
154   Blynk.run();
155   timer.run();
156 }
```

The simulation output shows the following data:

```
=====  
Cell 1: 2.76  
Cell 2: 2.80  
Cell 3: 3.14  
Cell 4: 2.98  
Health: Failure  
=====
```

STATUS –FAILURE- BLYNK

The Blynk Console interface shows a dashboard for an "Adaptive multicell battery". The dashboard displays the following data:

- CELL1: 2.76V
- CELL2: 2.8V
- CELL3: 3.14V
- CELL4: 2.98V
- AVERAGE VOLTAGE: 2.92V
- IMBALANCE PERCENTAGE: 13.02%
- HEALTH: Failure
- WEEKCELL: 1
- STRONGCELL: 3

STATUS –LOW- WOKWI (IF ANY ONE OF THE CELLS GOES BELOW 1.5V)

The Wokwi simulation interface displays a C++ code for a battery management system. The code includes functions for checking cell voltages, calculating average voltage, and triggering a low voltage warning. The simulation shows a 3D model of a battery pack with a red LED indicator and a green LCD screen displaying 'Avg: 2.23' and 'LOW VOLTAGE'. The console output shows the following data:

Cell	Voltage
Cell 1	0.00
Cell 2	2.80
Cell 3	3.14
Cell 4	2.98

Health: Failure
LOW VOLTAGE

STATUS –LOW- BLYNK

The Blynk Console interface displays a dashboard for an adaptive multicell battery. The dashboard includes several gauges and status indicators:

- CELL1: 0V
- CELL2: 2.8V
- CELL3: 3.14V
- CELL4: 2.98V
- AVERAGE VOLTAGE: 2.23V
- IMBALANCE PERCENTAGE: 100%
- HEALTH: LOW VOLTAGE!
- WEEKCELL: 1
- STRONGCELL: 3

STATUS –BALANCED/HEALTHY – WOKWI (IF ALL THE CELLS ARE HAVING SAME VOLTAGE)

The screenshot shows the Wokwi web-based IDE. On the left, the 'sketch.ino' file contains the following code:

```
126 void batteryEngine() {
127   if (lowVoltage || health == "Critical" || health == "Failure") {
128   } else {
129     digitalWrite(LED_PIN, LOW);
130     noTone(BUZZER);
131     digitalWrite(RELAY, HIGH);
132   }
133 }
134 // Blynk
135 Blynk.virtualWrite(V0, cell[0]);
136 Blynk.virtualWrite(V1, cell[1]);
137 Blynk.virtualWrite(V2, cell[2]);
138 Blynk.virtualWrite(V3, cell[3]);
139 Blynk.virtualWrite(V4, avgV);
140 Blynk.virtualWrite(V5, imbalance);
141
142 if (lowVoltage) {
143   Blynk.virtualWrite(V6, "LOW VOLTAGE!");
144 } else if (equalCells) {
145   Blynk.virtualWrite(V6, "Balanced/Healthy");
146 } else {
147   Blynk.virtualWrite(V6, health);
148 }
149
150 if (equalCells) {
151   Blynk.virtualWrite(V7, 0);
152   Blynk.virtualWrite(V8, 0);
153 } else {
154   String weakStr = "";
155 }
```

On the right, the 'Simulation' window shows a 3D model of a battery pack with four cells. Below the model, the following data is displayed:

```
Cell 1: 3.30
Cell 2: 3.30
Cell 3: 3.30
Cell 4: 3.30
Health: Healthy
```

STATUS –BALANCED/HEALTHY – BLYNK

The screenshot shows the Blynk Console interface for a project named 'Adaptive multicell battery'. The dashboard displays the following widgets:

- CELL1: 3.3V (0 to 4 scale)
- CELL2: 3.3V (0 to 4 scale)
- CELL3: 3.3V (0 to 4 scale)
- CELL4: 3.3V (0 to 4 scale)
- AVERAGE VOLTAGE: 3.3V (0 to 4 scale)
- IMBALANCE PERCENTAGE: 0% (0 to 100 scale)
- HEALTH: Balanced/Healthy
- WEEKCELL: 0
- STRONGCELL: 0

IN STRONG CELL AND WEEK CELL IT SHOWS ZERO - MEANS THAT ALL CELLS ARE HAVING THE SAME VOLTAGE.